

Rounak Ray

Curriculum Vitae

Office 310, Division of Applied Mathematics, Brown University
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📄 iamrounak-ray.github.io

Research Interests

Random graphs, Statistical physics, Interacting particle systems, Random walks, Random interface models, Large deviation

Work Experience

- 2025 - 2027 [Division of Applied Mathematics, Brown University](#) Post-doctoral research fellow
Jan - May 2025 [SLMath](#) (formerly **MSRI**) Post-doctoral research fellow

Education

- 2020–2024 **PhD**, [Eindhoven University of Technology](#).
Department: Mathematics & Computer Science
Supervisors: [Remco van der Hofstad](#), [Rajat Subhra Hazra](#)
Dissertation: [Stochastic Processes on Preferential Attachment Models](#)
2018–2020 **Master of Statistics (Hons.)**, [Indian Statistical Institute, Kolkata](#).
Specialisation: Probability Theory
2015–2018 **Bachelor of Statistics (Hons.)**, [Indian Statistical Institute, Kolkata](#).

Professional Experience

- November 2023 Organizer of "[Lecture series on interacting particle systems on random graphs](#)"
2021–2023 Teaching Assistant, Eindhoven University of Technology: Longitudinal Data Analysis
2022–2024 Teaching Assistant, Eindhoven University of Technology: Random Graphs

Ongoing Project

- Title **Finite dissociability of infinite graphs**
Collaborators [Kavita Ramanan](#)
Title **Bethe prediction for statistical physics models with non-pairwise interaction**
Collaborators [Kavita Ramanan](#), [I-Hsun Chen](#), [Janaky Murthy](#), [Bodhisatta Das](#), [Aritra Majumdar](#)
Title **Analysing CD behaviour in nano-lithography process** (Joint project with ASML)
Collaborators [Remco van der Hofstad](#), [B. Slachter](#) (ASML), [Haodong Zhu](#)

Publications

- 2026 [S. Bhamidi, Prof. Frank den Hollander, R. van der Hofstad, R. Ray](#) **Network evolution with self-reinforcement**. [ArXiv Preprint](#)
2026 [R. van der Hofstad, R. Ray](#) **Ising model on preferential attachment models**. [accepted in Journal of Statistical Physics](#)
2026 [R. Hazra, R. van der Hofstad, R. Ray](#). **Percolation on preferential attachment models**. [Electronic Journal of Probability, Volume 31 \(2026\)](#).
2026 [A. Garavaglia, R. Hazra, R. van der Hofstad, R. Ray](#). **Universality of the local limit of preferential attachment models**. [accepted in Annals of Applied Probability](#).
2025 [S. Banerjee, S. Bhamidi, R. van der Hofstad, R. Ray](#) **Non-equilibrium coagulation processes and subcritical percolation on evolving networks**. [ArXiv Preprint](#)

- 2023 [A. Cipriani, B. Dan, R. Hazra, R. Ray. Maximum of the Membrane Model on Regular Trees. Journal of Statistical Physics 190, Article number: 25\(2023\).](#)

Selected Talks

- 2026 Synergies between Geometry, Probability, and Computation in High Dimensions
2025 Probability and Statistics of Discrete Structures Seminar, SLMATH (formerly MSRI)
2024 International Indian Statistical Association Conference, Cochin, India.
2023 INFORMS APS-23 conference, Nancy, France
2023 Mark-Kac Seminar, Utrecht University, The Netherlands
2022 Spring School: Random Geometric Graph, TU Darmstadt, Germany.
2021 NETWORKS Training Week, Asperen, The Netherlands
2021 CNRS-IRP Bézout-Eurandom meeting, Institut Henri Poincaré, Paris, France.

Positions of Responsibilities

- 2021–2022 **Treasurer** of 59-th board of **Panache E.S.B.V**
2018–2019 **Treasurer** for **Integration 2018** and **Integration 2019**, the Annual Techno-Cultural-Sports Fest of **Indian Statistical Institute, Kolkata**
2017–2018 **Treasurer** of Hostel Mess run by a Students Body

Technical Skills

- Programming Languages PYTHON, C++
Software Packages R, $\text{\LaTeX}$

Extra Curriculars

- Hobbies Travelling, Nature Photography
Activities Drama, Football, Chess, Bridge

Languages

- Bengali Mother Tongue
Hindi Native Language
English 2nd Language